

ECO-HABITAT HEALTH TRACKER 🌳

A MINI-PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project “**ECO-HABITAT HEALTH TRACKER**” is the bonafide work of “**KARTHAVYA S, ILAKKIYA P**” who carried out the project work under my supervision.

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This mini project report is submitted for the viva voce examination to be held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

Habitat Tracker is a Java Swing application that monitors wildlife reserves by tracking environmental factors and evaluating risk levels. Users can enter habitat data like temperature, humidity, and pollution readings manually. The system saves this information with timestamps for easy record-keeping. It also provides graphical displays of temperature trends, allowing users to see changes over time. A detailed habitat list and risk-level display help spot potential threats and prompt timely responses. With its interactive interface and data-based insights, Habitat Tracker aids in effective monitoring and proactive conservation of wildlife environments.

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1. KARTHAVYA S

2. ILAKKIYA P

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CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The introduction page presents an overview of the Habitat Tracker system. It highlights its purpose—monitoring wildlife reserves through parameter readings, visual graphs, and risk assessments. Users can easily navigate to view habitats, record data, and track environmental changes in real time.

1.2 SCOPE OF THE WORK

The Habitat Tracker project focuses on monitoring wildlife habitats by recording environmental parameters, displaying habitat lists, and assessing risk levels. It enables manual data entry with timestamped readings and graphical analysis. The system supports efficient conservation management and can be expanded for real-time sensor-based monitoring.

1.3 PROBLEM STATEMENT

Wildlife habitats face increasing threats from climate changes, pollution, and human interference, making constant monitoring essential. Traditional methods of tracking environmental conditions are time-consuming and inefficient. There is a need for a digital system that records, analyzes, and alerts risk levels to ensure effective habitat protection and management.

1.4 AIM AND OBJECTIVES OF THE PROJECT

To develop a Java Swing-based application that monitors wildlife habitats by tracking environmental parameters, assessing risk levels, and supporting proactive conservation management.

Objectives:

- To record and store environmental readings with timestamps.
- To display habitat lists and their respective risk levels.
- To visualize temperature data through graphs.
- To enable manual data entry and secure database management.
- To generate alerts for abnormal or risky habitat conditions

CHAPTER 2

SYSTEM SPECIFICATIONS

2.1 HARDWARE SPECIFICATIONS

Processor	:	Intel i5
Memory Size	:	16GB (Minimum)

HDD	:	64GB (Minimum)
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2.2 SOFTWARE SPECIFICATIONS

Operating System	:	WINDOWS 11
Front – End	:	Java Swing
Back - End	:	MySql
Language	:	Java,SQL

CHAPTER 3

MODULE DESCRIPTION

This application consists of a single module. When the program runs, it will display a login window asking for confirmation. The person who interacts can log in as an **Administrator**. The description of the module is as follows:

Admin Login

When the person logs in as an Administrator, they must enter a valid username and password. The administrator has full control over the system, including adding, updating, and deleting habitat details. They can manually enter environmental readings such as temperature, humidity, and pollution

levels, view risk levels, analyze graphical trends, and manage the database efficiently.

CHAPTER 4

SAMPLE CODING 1

This section represents the Main Java class that serves as the entry point of the Habitat Tracker application. It initializes the system, launches the login window, and connects the interface with the backend modules such as database handling and habitat monitoring functions.

Main.java

```
import javax.swing.*;

public class Main {

    public static void main(String[] args) {

        SwingUtilities.invokeLater(() -> {

            LoginDialog loginDlg = new LoginDialog(null);

            loginDlg.setVisible(true);

            if (loginDlg.isSucceeded()) {

                new HabitatQualityUI(); // Launch main JSwing UI

            } else {

                System.exit(0); // Exit if login fails or canceled

            }

        });
    }
}
```

```

    }

}

```

SAMPLE CODING 2

This section represents the Login Page module, which handles user authentication for the Habitat Tracker application. It verifies the administrator's username and password, ensuring secure access to the system. Upon successful login, the main dashboard is launched; otherwise, an error message is displayed prompting re-entry of valid credentials.

Login.java

```

package src.view;

import javax.swing.*.*;

import java.awt.*.*;

public class LoginDialog extends JDialog {

    private boolean succeeded;

    public LoginDialog(Frame parent) {

        super(parent, "Administrator Login", true);

        // Main panel with padding

        JPanel panel = new JPanel(new GridBagLayout());

        panel.setBorder(BorderFactory.createEmptyBorder(20, 20, 20, 20));

        GridBagConstraints gbc = new GridBagConstraints();

        gbc.insets = new Insets(10, 10, 10, 10);

```

```
gbc.fill = GridBagConstraints.HORIZONTAL;

// Username

gbc.gridx = 0;

gbc.gridy = 0;

panel.add(new JLabel("Username:"), gbc);

gbc.gridx = 1;

JTextField tfUsername = new JTextField(20);

panel.add(tfUsername, gbc);

// Password

gbc.gridx = 0;

gbc.gridy = 1;

panel.add(new JLabel("Password:"), gbc);

gbc.gridx = 1;

JPasswordField pfPassword = new JPasswordField(20);

panel.add(pfPassword, gbc);

// Buttons

gbc.gridx = 0;

gbc.gridy = 2;

JButton btnCancel = new JButton("Cancel");
```

```
panel.add(btnCancel, gbc);

gbc.gridx = 1;

JButton btnLogin = new JButton("Login");

panel.add(btnLogin, gbc);

getContentPane().add(panel, BorderLayout.CENTER);

// Button actions

btnLogin.addActionListener(e -> {

    String username = tfUsername.getText().trim();

    String password = new String(pfPassword.getPassword()).trim();

    if ("admin".equals(username) && "admin123".equals(password)) {

        succeeded = true;

        dispose();

    } else {

        JOptionPane.showMessageDialog(LoginDialog.this,

            "Invalid username or password",

            "Login Failed",

            JOptionPane.ERROR_MESSAGE);

        tfUsername.setText("");

        pfPassword.setText("");

    }

});
```

```
succeeded = false;  
  
} };
```

CHAPTER 5

SCREEN SHOTS

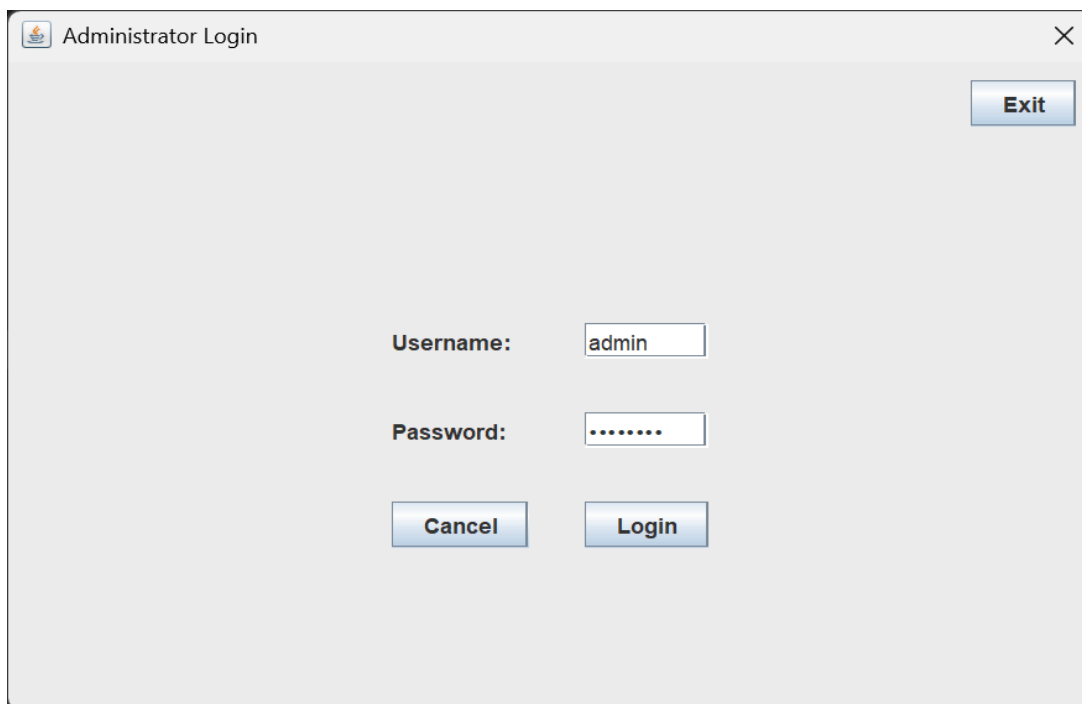


Fig 5.1 Introduction page



Fig 5.2 Habitats list

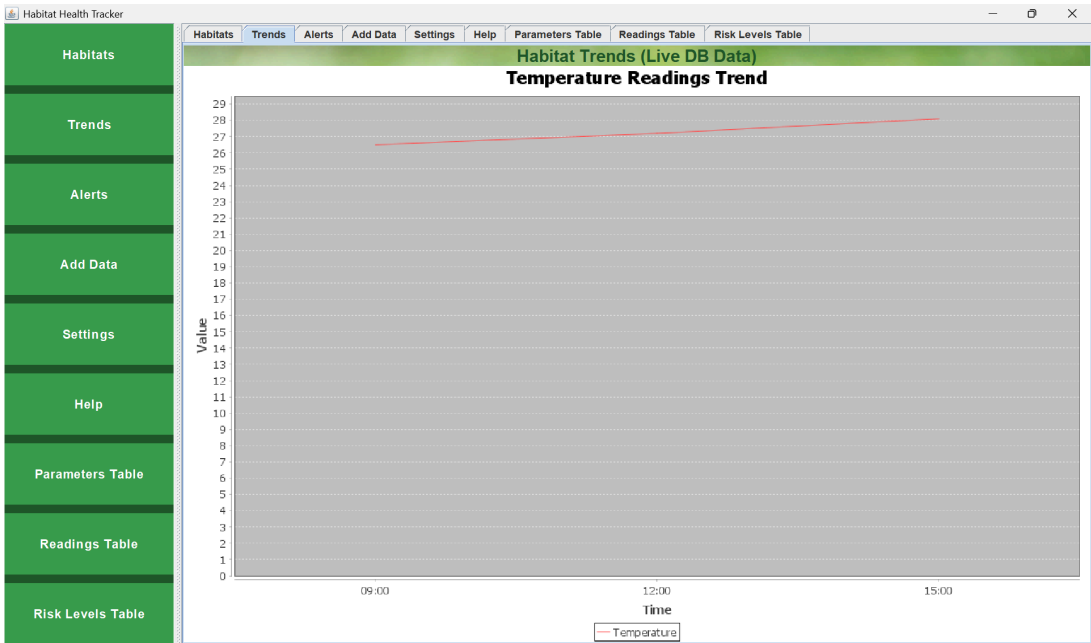


Fig 5.3 Trend Analysis

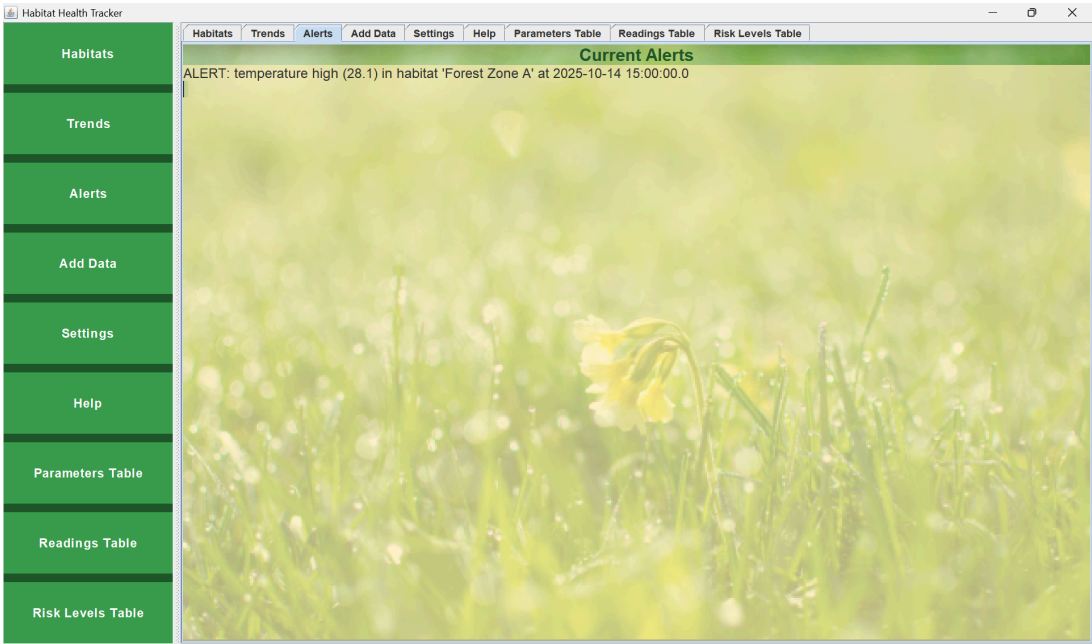


Fig 5.4 Alerting System

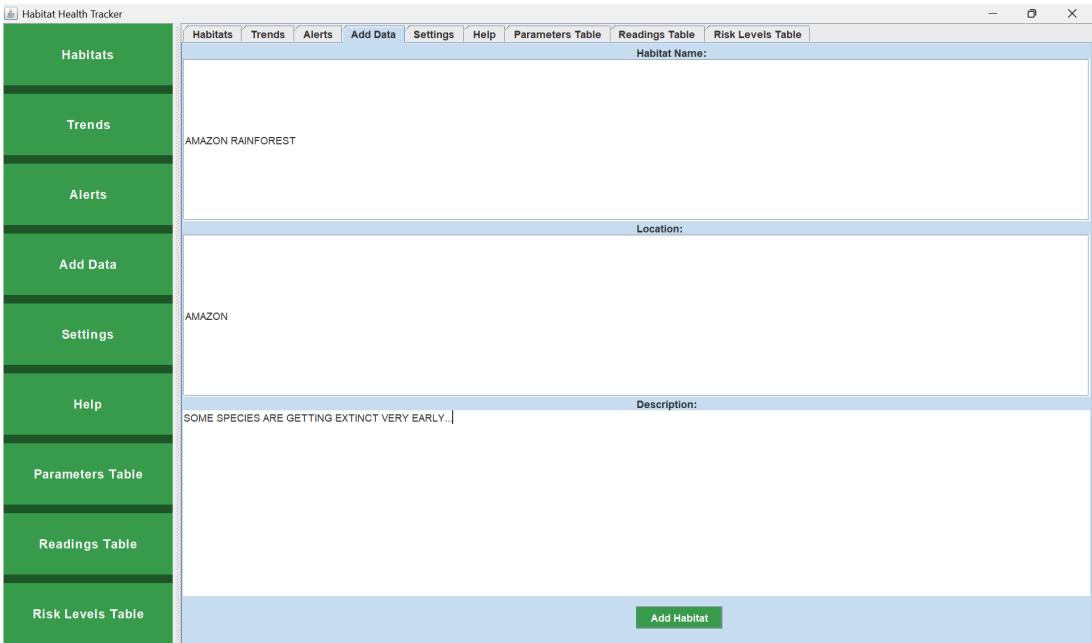


Fig 5.5 Data Addition

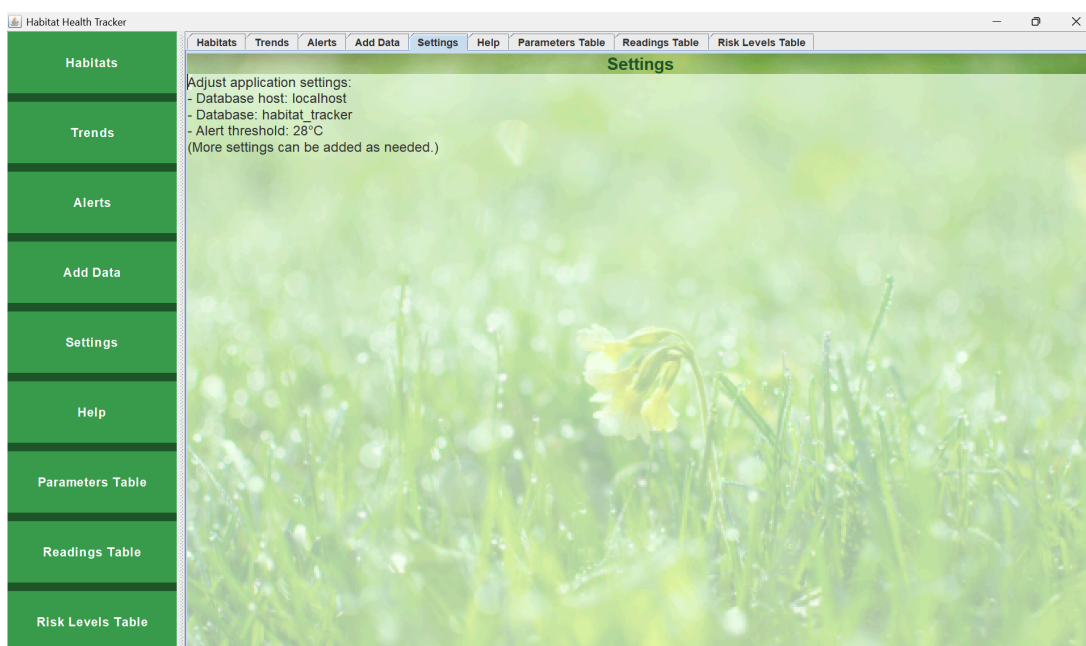


Fig 5.6 Settings

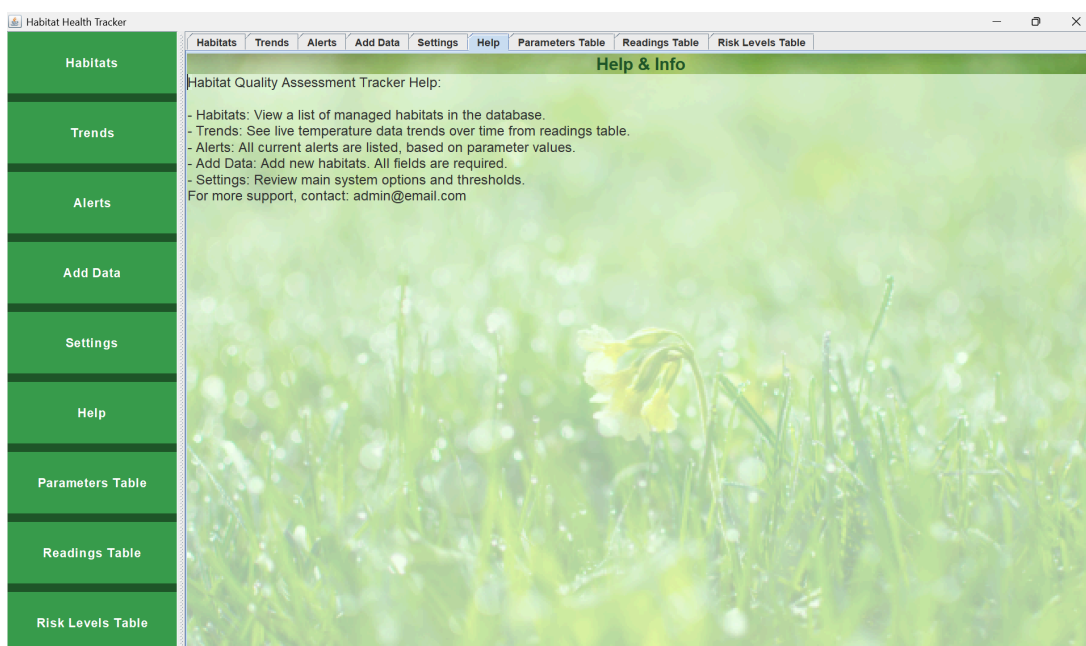


Fig 5.7 Help Tab

The screenshot shows the 'Habitat Health Tracker' application window. On the left is a green sidebar with buttons for 'Habitats', 'Trends', 'Alerts', 'Add Data', 'Settings', 'Help', 'Parameters Table', 'Readings Table', and 'Risk Levels Table'. The 'Parameters Table' button is selected. The main area displays a table titled 'Table: parameters' with two columns: 'parameter_id' and 'name'. The table contains six rows of data.

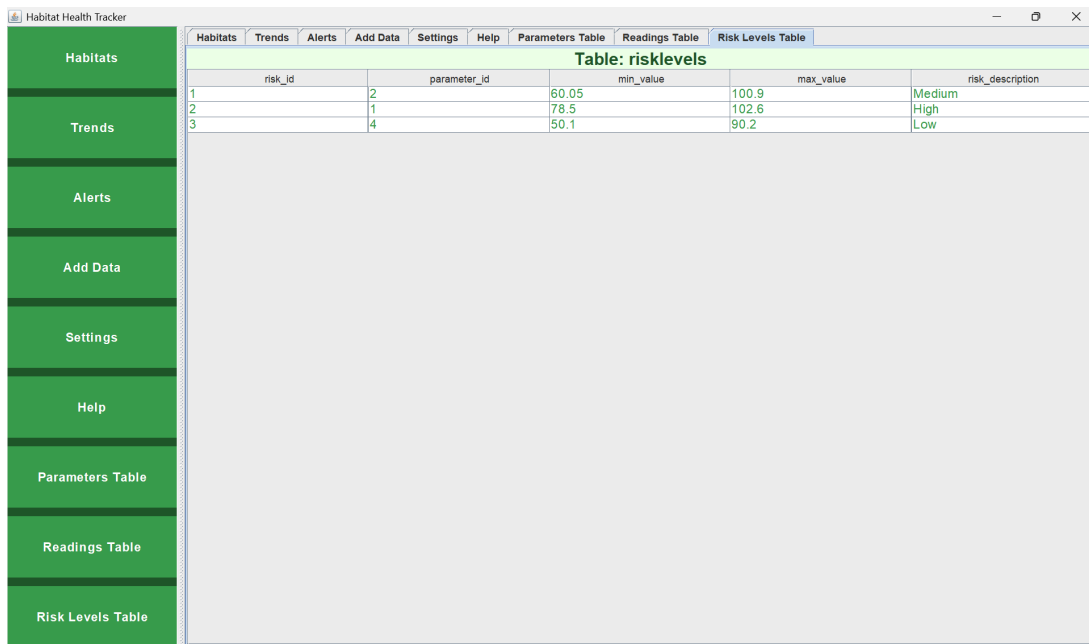
parameter_id	name
1	temperature
2	Temperature
3	Humidity
4	Soil Moisture
5	Light Intensity
6	pH Level

Fig 5.8 Parameters Table

The screenshot shows the 'Habitat Health Tracker' application window with the 'Readings Table' button selected in the sidebar. The main area displays a table titled 'Table: readings' with five columns: 'reading_id', 'habitat_id', 'parameter_id', 'value', and 'timestamp'. The table contains three rows of data.

reading_id	habitat_id	parameter_id	value	timestamp
1	1	1	26.5	2025-10-14T09:00
2	1	1	27.2	2025-10-14T12:00
3	1	1	28.1	2025-10-14T15:00

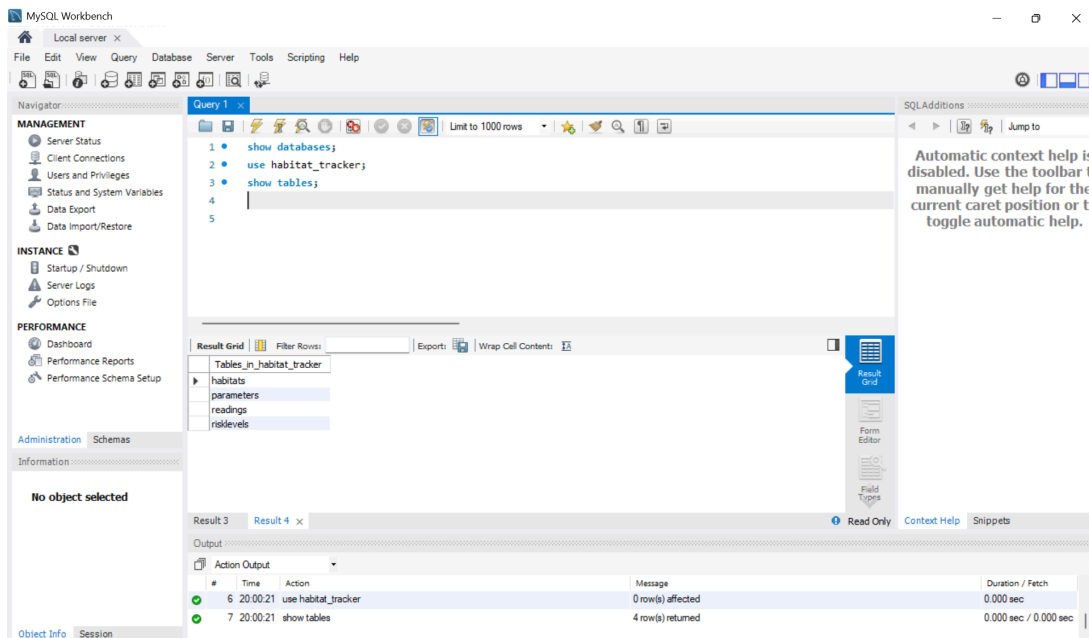
Fig 5.9 Readings Table



The screenshot shows the 'Habitat Health Tracker' application with a sidebar menu on the left containing: Habitats, Trends, Alerts, Add Data, Settings, Help, Parameters Table, Readings Table, and Risk Levels Table. The 'Risk Levels Table' is selected, displaying a table with the following data:

Table: risklevels				
risk_id	parameter_id	min_value	max_value	risk_description
1	2	60.05	100.9	Medium
2	1	78.5	102.6	High
3	4	50.1	90.2	Low

Fig 5.10 Risk Levels Table



The screenshot shows the MySQL Workbench interface. The 'Query' tab is active, displaying the following SQL queries:

```

1 show databases;
2 use habitat_tracker;
3 show tables;
4
5

```

The 'Result Grid' shows the output of the queries:

Tables_in_habitat_tracker
habitats
parameters
readings
risklevels

The 'Output' tab shows the execution results:

#	Time	Action	Message	Duration / Fetch
6	20:00:21	use habitat_tracker	0 row(s) affected	0.000 sec
7	20:00:21	show tables	4 row(s) returned	0.000 sec / 0.000 sec

Fig 5.11 Database Creation(Tables)

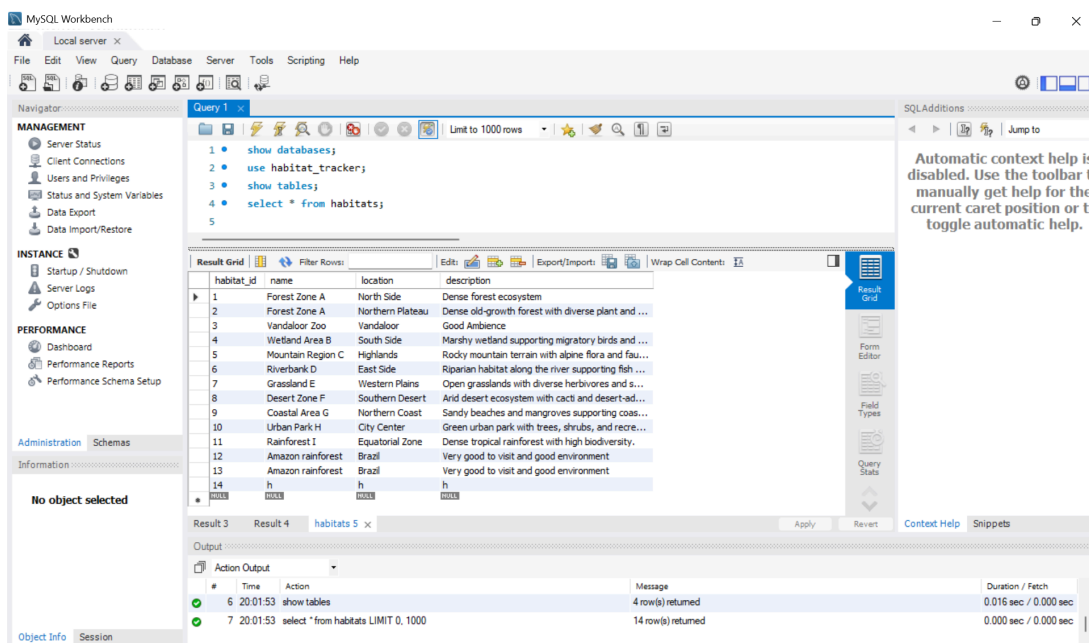


Fig 5.12 Database Creations(Values)

CHAPTER 6

CONCLUSION AND ENHANCEMENTS

The Habitat Tracker project successfully demonstrates how technology can aid wildlife conservation by monitoring environmental parameters and assessing habitat risk levels. The system provides an easy-to-use interface for administrators to record, view, and analyze data effectively. It ensures timely awareness of changes through readings and visual graphs. Future enhancements can include integrating IoT sensors for automatic data collection, real-time cloud synchronization, and

mobile application support. Additional AI-based analysis could also be implemented to predict potential threats and provide preventive alerts, making the system more intelligent and impactful for large-scale habitat management.

REFERENCES

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